BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Preventing Network Attacks

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the concept of SQL injection?

2. What does a firewall control?

A) A monitor's brightness
B) Allowed network traffic
C) All files' storage size
D) The CPU's clock speed

3. "HTTP is the protocol used only to send email to a mail server." Correct this. Explain why.

4. A pupil logs in to a website. "HTTPS proves every website is honest." Correct this. What does HTTPS protect?

Today's objectives

Explain the core ideas of preventing network attacks using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Firewall
Filters network traffic according to predefined rules.

Encryption
Makes data unreadable without the appropriate key.

Authentication
Confirms identity; MFA uses multiple factors.

Updates and backups
Patching reduces exploitable vulnerabilities; recoverable backups reduce loss after incidents.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Firewall	Filters network traffic according to predefined rules.
Encryption	Makes data unreadable without the appropriate key.
Authentication	Confirms identity; MFA uses multiple factors.
Updates and backups	Patching reduces exploitable vulnerabilities; recoverable backups reduce loss after incidents.

I Do - One worked example

A school wants remote access for staff while protecting student data. Recommend layered controls.

1. Require authenticated accounts and MFA.
2. Use encrypted connections to protect data in transit.
3. Filter unnecessary traffic using appropriate firewall policies.
4. Apply updates and maintain tested backups; explain different risk each control addresses.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A company uses only passwords, with no MFA. Explain an improvement.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change the AND condition so both checks are required before access is permitted.

```
passwordOK = true
secondFactorOK = false
if passwordOK OR secondFactorOK then
print("Allow")
else
print("Deny")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. Explain why encryption protects intercepted information. [3 marks]

Q3. State one role of a firewall. [2 marks]

Q4. Explain why software updates can improve security. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Encryption alone guarantees nobody will ever steal a password. ☐ T ☐ F
2. Filters network traffic according to predefined rules. ☐ T ☐ F
3. Makes data unreadable without the appropriate key. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Unsafe input changes an SQL query. An attacker may then view or change data. ☐

Q2 B - Allowed network traffic ☐

Q3 Transferring web resources between web clients and servers. ☐

Q4 HTTPS encrypts data in transit. It does not prove the website or its owner is honest. ☐

Code First - Model

```
passwordOK = true
secondFactorOK = false
if passwordOK AND secondFactorOK then
print("Allow")
else
print("Deny")
endif
```

Output: Deny

We Do - Check the model

1. Require user training for phishing-resistant practice. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 The interceptor sees ciphertext; without the relevant key it is not readily readable. ☐

Q3 Allow or block traffic according to filtering rules. ☐

Q4 Fix known vulnerabilities that might otherwise be exploited. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: